
Midterm Exam Problem 2 - Nouh Shaikh

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% Bisection roots method was used by the file provided on brightspace.
% Secant roots was used from the previous problem.
clear; clc; close all;

func_1 = @(x) 1.5 * x.^3 .* sin(0.4 * x.^2) - 3 * exp(-x + 3) + 2 * x .*
log(x);

xb = [];
x = 1e-5;
dx = 0.1;

while x + dx <= 5
    if func_1(x) * func_1(x + dx) < 0
        xb = [xb; x, x + dx];
    end
    x = x + dx;
end

fprintf('Matrix of Brackets (xb):\n'); disp(xb);
num_roots = size(xb, 1);

figure(1); hold on; grid on; xlabel('x'); ylabel('f(x)'); title('Figure 1:
Bisection');
fplot(func_1, [1, 5], 'b', 'LineWidth', 1.5);
fplot(@(x) 0, [1, 5], 'r--', 'LineWidth', 0.5);

roots_bisection = zeros(num_roots, 1);
for i = 1:num_roots
    roots_bisection(i) = Bisection_roots(func_1, xb(i,1), xb(i,2));
    plot(roots_bisection(i), 0, 'or', 'MarkerSize', 8);
end
fprintf('Roots via Bisection:\n'); disp(roots_bisection);

figure(2); hold on; grid on; xlabel('x'); ylabel('f(x)'); title('Figure 2:
fzero');
fplot(func_1, [1, 5], 'b', 'LineWidth', 1.5);
fplot(@(x) 0, [1, 5], 'r--', 'LineWidth', 0.5);

roots_fzero = zeros(num_roots, 1);
for i = 1:num_roots
    roots_fzero(i) = fzero(func_1, (xb(i,1) + xb(i,2)) / 2);
    plot(roots_fzero(i), 0, 'ks', 'MarkerFaceColor', 'k', 'MarkerSize', 5);
end
fprintf('Roots via fzero:\n'); disp(roots_fzero);

figure(3); hold on; grid on; xlabel('x'); ylabel('f(x)'); title('Figure 3:
Secant');
fplot(func_1, [1, 5], 'b', 'LineWidth', 1.5);
fplot(@(x) 0, [1, 5], 'r--', 'LineWidth', 0.5);
```

```

roots_secant = zeros(num_roots, 1);
for i = 1:num_roots
    roots_secant(i) = Secant_roots(func_1, xb(i,1), xb(i,2));
    plot(roots_secant(i), 0, 'bd', 'MarkerFaceColor', 'b', 'MarkerSize', 5);
end
fprintf('Roots via Secant:\n'); disp(roots_secant);

```

Matrix of Brackets (xb):

1.7000	1.8000
2.8000	2.9000
3.9000	4.0000
4.8000	4.9000

Warning: Function behaves unexpectedly on array inputs. To improve performance, properly vectorize your function to return an output with the same size and shape as the input arguments.

iter	xr	err
1	1.75	0.028571
2	1.78	0.014084
3	1.79	0.006993
4	1.78	0.003509
5	1.78	0.001757
6	1.78	0.000878
7	1.78	0.000439
8	1.78	0.000219
9	1.78	0.000110
10	1.78	0.000055

iter	xr	err
1	2.85	0.017544
2	2.83	0.008850
3	2.84	0.004405
4	2.83	0.002207
5	2.83	0.001103
6	2.83	0.000552
7	2.83	0.000276
8	2.83	0.000138
9	2.83	0.000069
10	2.83	0.000034

iter	xr	err
1	3.95	0.012658
2	3.93	0.006369
3	3.94	0.003175
4	3.93	0.001590
5	3.93	0.000796
6	3.93	0.000398
7	3.93	0.000199
8	3.93	0.000099
9	3.93	0.000050
10	3.93	0.000025

iter	xr	err
1	4.85	0.010309
2	4.88	0.005128
3	4.89	0.002558

4	4.88	0.001280
5	4.88	0.000641
6	4.88	0.000320
7	4.88	0.000160
8	4.88	0.000080
9	4.88	0.000040
10	4.88	0.000020

Roots via Bisection:

1.7812
2.8332
3.9299
4.8762

Warning: Function behaves unexpectedly on array inputs. To improve performance, properly vectorize your function to return an output with the same size and shape as the input arguments.

Roots via fzero:

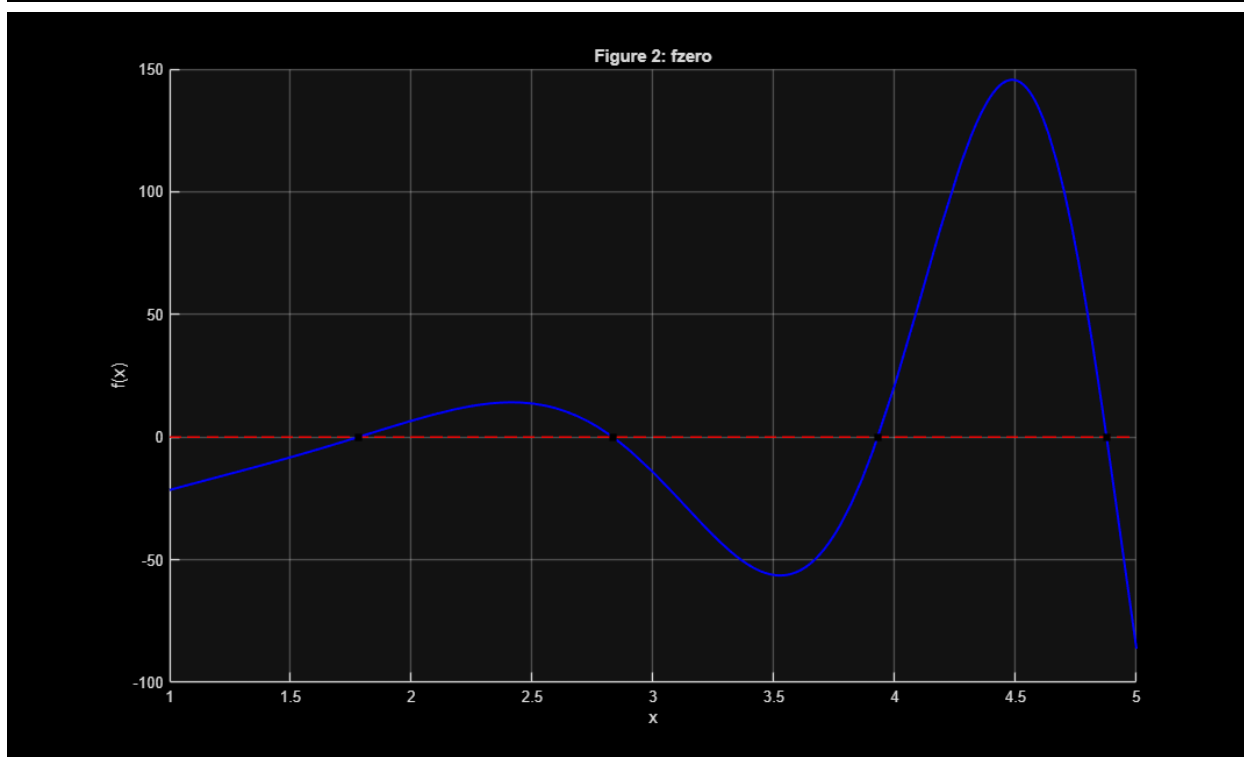
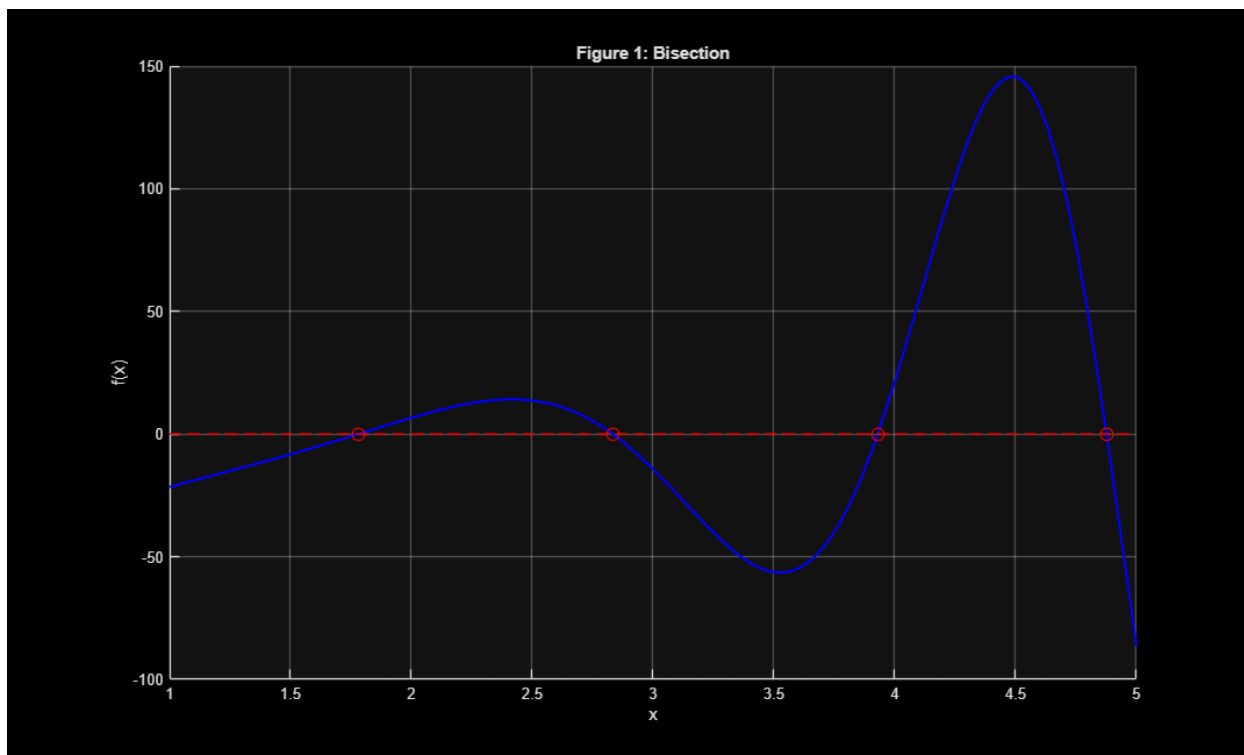
1.7812
2.8332
3.9300
4.8762

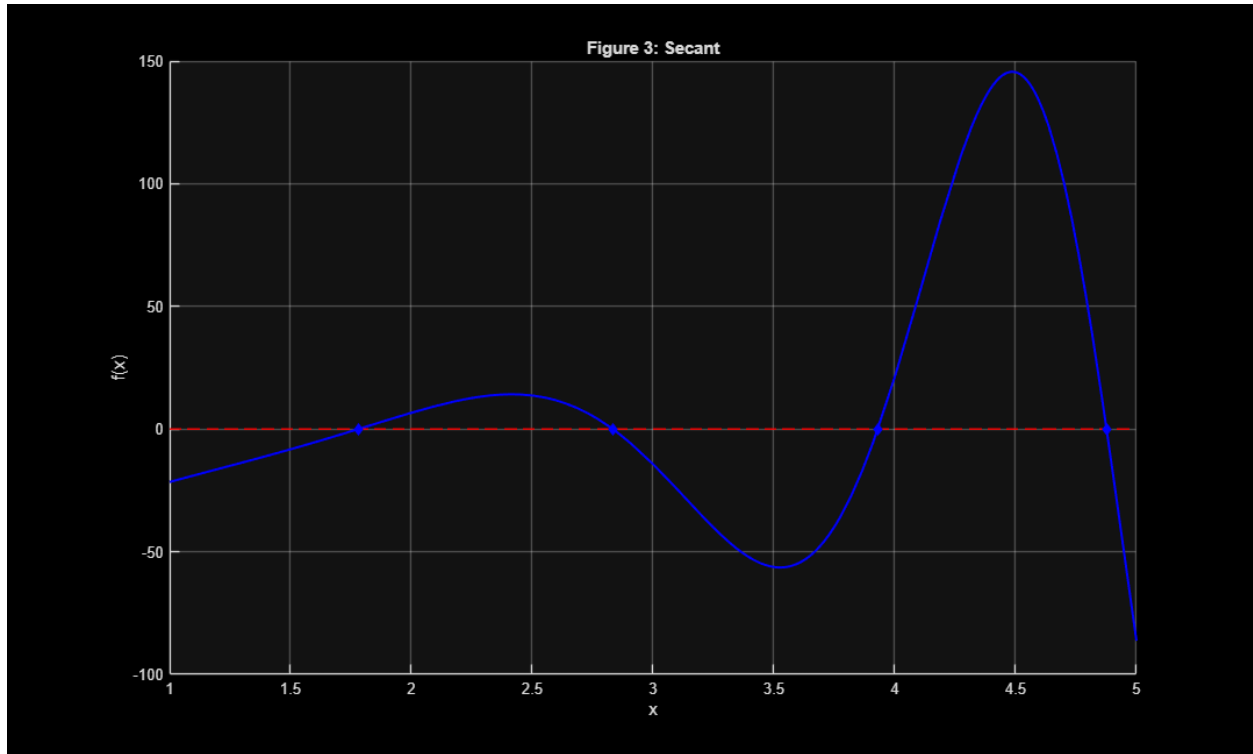
Warning: Function behaves unexpectedly on array inputs. To improve performance, properly vectorize your function to return an output with the same size and shape as the input arguments.

Iter: 1, xr = 1.7810, Error = 1.0653e-02
Iter: 2, xr = 1.7812, Error = 7.3746e-05
Iter: 1, xr = 2.8305, Error = 2.4559e-02
Iter: 2, xr = 2.8330, Error = 8.6848e-04
Iter: 3, xr = 2.8332, Error = 7.2297e-05
Iter: 1, xr = 3.9279, Error = 1.8348e-02
Iter: 2, xr = 3.9298, Error = 4.8165e-04
Iter: 3, xr = 3.9300, Error = 3.2658e-05
Iter: 1, xr = 4.8750, Error = 5.1346e-03
Iter: 2, xr = 4.8762, Error = 2.5558e-04
Iter: 3, xr = 4.8762, Error = 2.9940e-06

Roots via Secant:

1.7812
2.8332
3.9300
4.8762





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